

## OVERVIEW

This assignment covers the Evaluation phase of the Information System Research (ISR) process (Figure 1). You will need to select key conceptual and system artefacts and evaluate whether your artefacts solve the stated research problems. The evaluation methods that you use are based on the taxonomy of Hevner et al. (2004) (see Table 1). You will also need to assess system artefact quality using the FURPS model: Functionality, Usability, Reliability, Performance, and Supportability (Grady & Caswell, 1987), as covered in lectures.

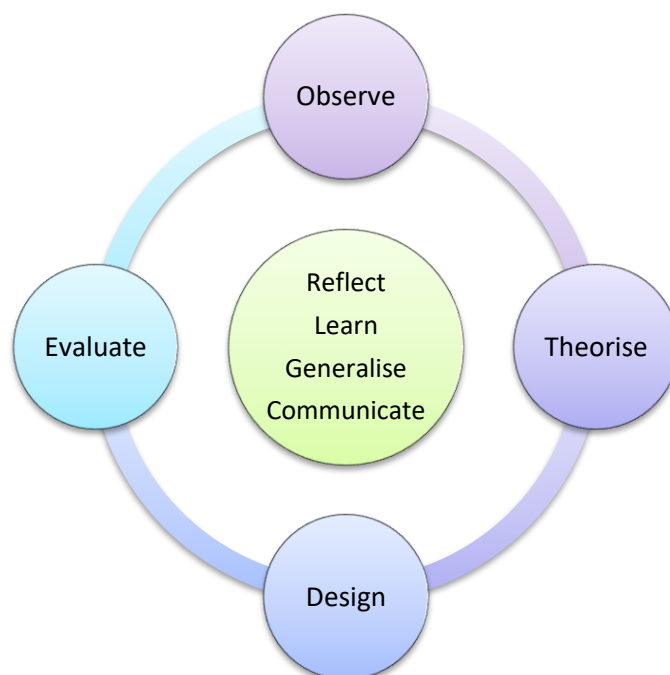


Figure 1: Information System Research Process

**Table 1. Evaluation Methods (Hevner et al., 2004)**

|                  |                                                                                                                                                 |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Observational | Case Study – Study artifact in depth in business environment                                                                                    |
|                  | Field Study – Monitor use of artifact in multiple projects                                                                                      |
| 2. Analytical    | Static Analysis – Examine structure of artifact for static qualities (e.g., complexity)                                                         |
|                  | Architecture Analysis – Study fit of artifact into technical IS architecture                                                                    |
|                  | Optimization – Demonstrate inherent optimal properties of artifact or provide optimality bounds on artifact behavior                            |
|                  | Dynamic Analysis – Study artifact in use for dynamic qualities (e.g., performance)                                                              |
| 3. Experimental  | Controlled Experiment – Study artifact in controlled environment for qualities (e.g., usability)                                                |
|                  | Simulation – Execute artifact with artificial data                                                                                              |
| 4. Testing       | Functional (Black Box) Testing – Execute artifact interfaces to discover failures and identify defects                                          |
|                  | Structural (White Box) Testing – Perform coverage testing of some metric (e.g., execution paths) in the artifact implementation                 |
| 5. Descriptive   | Informed Argument – Use information from the knowledge base (e.g., relevant research) to build a convincing argument for the artifact's utility |
|                  | Scenarios – Construct detailed scenarios around the artifact to demonstrate its utility                                                         |

The evaluation must connect back to the requirements and the problem articulated in PIRQOA. This assignment contributes directly to your final research paper, submitted in Assignment 9.

## DELIVERABLES

Submit your assignment in **.docx format** via Canvas by **11:59 AM, Friday, 2<sup>nd</sup> October**. Late submissions will incur penalties as per university policy. Extensions are only granted under force majeure conditions.

## GUIDELINES AND REQUIREMENTS

- Refer to the marking rubric for guidance.
- Ensure your document is professional, well-structured, and free of spelling/grammar errors.
- Use APA 7 referencing style throughout.
- Label all tables and figures consistently.
- Formatting Instructions
  - If you have selected a conference or journal outlet for publication, then format your document according to the official template and style guide of that venue. Include the name of the selected venue in the header or cover page.
  - If you have not selected a journal or conference, use the following default format: Font: Times New Roman, Title: 16pt, Headings: 14pt, Body Text: 12pt, Labels: 11pt, Table/Figure Text: 10pt, Line Spacing: 1.5.

## ASSIGNMENT STRUCTURE AND MARKING SCHEME

| Section                               | Step | Task                                                                                                                       | Hint                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Weight (%) | Suggested Word Count                                    |
|---------------------------------------|------|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------------------------------------------------------|
| <b>Title, Abstract, and Keywords</b>  | 1    | Revise the title, abstract, and keywords as necessary.                                                                     | <ul style="list-style-type: none"> <li>Update the title if required.</li> <li>Update the abstract. It should summarise the research context, insights from A1, and the artefacts developed in this phase.</li> <li>Update the keywords if necessary.</li> </ul> <p>These updates might be required as you progress with the <b>SLR</b>, <b>SSR</b>, and <b>PIRQOA</b>.</p>                                                                                                                                                               | 1 %        | Title ≤ 10 words<br>Abstract ≤150 words<br>Keywords ≤ 5 |
| <b>Conceptual Artefact Evaluation</b> | 2    | Evaluate <b>key conceptual artefact(s)</b> using evaluation methods from Hevner et al. (2004).                             | <p>Select <b>at least one</b> conceptual artefact that makes an important contribution to your research, and evaluate it using the methods below:</p> <ol style="list-style-type: none"> <li><b>Observational:</b> GenAI Interview</li> <li><b>SLR:</b> Academic Literature</li> <li><b>Descriptive:</b> Informed Argument and Scenarios</li> <li><b>Observational:</b> Human Expert Interview (1% Bonus mark)</li> </ol>                                                                                                                | 5.5 %      | 800-1200                                                |
| <b>Design Artefact Evaluation</b>     | 3    | Evaluate the <b>executable proof-of-concept app/game</b> using FURPS and the evaluation methods from Hevner et al. (2004). | <p>Evaluate the <b>executable proof-of-concept app/game</b> as one of the key system artefacts using the methods below:</p> <ol style="list-style-type: none"> <li><b>FURPS</b> (F and U)</li> <li><b>Analytical:</b> Static, Dynamic and Optimisation</li> <li><b>Experimental:</b> Simulation</li> <li><b>Testing:</b> Black Box and White Box</li> <li><b>Descriptive:</b> Informed Argument and Scenarios</li> <li><b>SLR:</b> Academic Literature</li> <li><b>Observational:</b> Human Expert Interview (1% Bonus mark )</li> </ol> | 5.5 %      | 800-1200                                                |

| Section                               | Step | Task                                                                                                     | Hint                                                                                                                                                                | Weight (%) | Suggested Word Count |
|---------------------------------------|------|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------------------|
| <b>Results &amp; interpretation</b>   | 4    | Present and interpret results. Show the evaluation tests whether the artefact solves the stated problem. | Use tables and figures to interpret what the results do and do not show.<br>Map each evaluation result back to the requirements and questions stated in the PIRQOA. | 2 %        | 500–750              |
| <b>Presentation &amp; Referencing</b> | 5    | Presentation & Structure.                                                                                | Formatting, clarity, flow, and visual layout.                                                                                                                       | 0.5 %      | -                    |
|                                       | 6    | Referencing & Academic Integrity.                                                                        | Correct citation style, academic tone, and originality.                                                                                                             | 0.5 %      | -                    |

## REFERENCES

- Hevner, A. R., March, S. T., Park, J., & Ram, S. (2004). Design science in information systems research. *MIS Quarterly*, 28(1), 75–105. <https://doi.org/10.2307/25148625>
- Grady, R., & Caswell, D. (1987). *Software Metrics: Establishing a Company-wide Program*. Prentice Hall. pp. 159. ISBN 0-13-821844-7.